

## Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO: 2520030071

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Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS dd1_db;  
USE dd1_db;
```

Output: -

|   |    |          |                                      |                   |           |
|---|----|----------|--------------------------------------|-------------------|-----------|
| ✓ | 39 | 15:09:29 | CREATE DATABASE IF NOT EXISTS dd1_db | 1 row(s) affected | 0.110 sec |
| ✓ | 40 | 15:09:34 | USE dd1_db                           | 0 row(s) affected | 0.015 sec |

Query 2: -DDL (DATA DEFINITION LANGUAGE)

CREATE TABLE

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```
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE  
);
```

Output: -

|   |    |          |                                                  |                   |           |
|---|----|----------|--------------------------------------------------|-------------------|-----------|
| ✓ | 41 | 15:12:39 | CREATE TABLE bank_transactions ( txn_id INT P... | 0 row(s) affected | 0.203 sec |
|---|----|----------|--------------------------------------------------|-------------------|-----------|

Query 3: -ALTER TABLE ADD COLUMN

-- G. Hemanth Venkatesh

```
ALTER TABLE bank_transactions  
ADD account_no VARCHAR(20);
```

Output: -

|   |    |          |                                                  |                   |                                                |
|---|----|----------|--------------------------------------------------|-------------------|------------------------------------------------|
| ✓ | 42 | 15:14:01 | ALTER TABLE bank_transactions ADD account_no ... | 0 row(s) affected | Records: 0 Duplicates: 0 Warnings: 0 0.093 sec |
|---|----|----------|--------------------------------------------------|-------------------|------------------------------------------------|

Query 4: -ALTER TABLE MODIFY COLUMN

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```
ALTER TABLE bank_transactions  
MODIFY customer_name VARCHAR(100);
```

Output: -

✓ 43 15:15:05 ALTER TABLE bank\_transactions MODIFY customer\_name VARCHAR(...) 0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0 0.109 sec

Query 5: -RENAME TABLE

```
RENAME TABLE  
bank_transactions TO customer_transactions;
```

Output: -

✓ 44 15:17:35 RENAME TABLE bank\_transactions TO customer\_transactions 0 row(s) affected 0.063 sec

Query 6: -TRUNCATE TABLE

```
TRUNCATE TABLE customer_transactions;
```

Output: -

✓ 46 15:19:47 TRUNCATE TABLE customer\_transactions 0 row(s) affected 0.032 sec

Query 7: -DROP TABLE

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```
DROP TABLE IF EXISTS bank_backup;
```

Output: -

⚠ 48 15:21:33 DROP TABLE IF EXISTS bank\_backup 0 row(s) affected, 1 warning(s): 1051 Unknown table '...' 0.109 sec

Query 8: - DML (DATA MANIPULATION LANGUAGE)

INSERT SAMPLE DATA

```

1 ● ○ CREATE TABLE customers_transactions (
2     txn_id INT PRIMARY KEY,
3     customer_name VARCHAR(100),
4     branch_name VARCHAR(50),
5     transaction_type VARCHAR(20),
6     amount DECIMAL(10,2),
7     transaction_date DATE
8 );
9 ● INSERT INTO customers_transactions VALUES
0     (101, 'G. Hemanth Venkatesh', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
1     (102, 'G. Hemanth Venkatesh', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
2     (103, 'Akshara Reddy', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
3     (104, 'Anirudh Reddy', 'Vizag', 'Deposit', 8000, '2024-01-10'),
4     (105, 'Yashwanth', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
5     (106, 'Adithya Vardhan', 'Vizag', 'Deposit', 15000, '2024-01-12'),
6     (107, 'Ananaya Varshitha', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
7     (108, 'Varnika Guptha', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
8     (109, 'Prerka', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
9     (110, 'Nihita', 'Vijayawada', 'Deposit', 11000, '2024-01-16');

```

Output: -

|   |    |          |                                                   |                                                       |           |
|---|----|----------|---------------------------------------------------|-------------------------------------------------------|-----------|
| ✓ | 52 | 15:27:20 | CREATE TABLE customers_transactions ( txn_id l... | 0 row(s) affected                                     | 0.031 sec |
| ✓ | 53 | 15:27:28 | INSERT INTO customers_transactions VALUES (101... | 10 row(s) affected Records: 10 Duplicates: 0 Warni... | 0.015 sec |

Query 9: -INSERT

```

2520030071-G. Hemanth Venkatesh
INSERT INTO customers_transactions
VALUES
(111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');

```

Output: -

|   |    |          |                                                   |                   |           |
|---|----|----------|---------------------------------------------------|-------------------|-----------|
| ✓ | 54 | 15:28:57 | INSERT INTO customers_transactions VALUES (111... | 1 row(s) affected | 0.062 sec |
|---|----|----------|---------------------------------------------------|-------------------|-----------|

Query 10: -UPDATE

```

UPDATE customers_transactions
SET amount = 5000
WHERE txn_id = 105;

```

Output: -

56 15:30:53 UPDATE customers\_transactions SET amount = 5000 WHERE tx... 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

Query 11: -DELETE

0.015 sec

```
DELETE FROM customers_transactions
WHERE txn_id = 111;
```

Output: -

58 15:32:09 DELETE FROM customers\_transactions WHERE txn\_id = 111 1 row(s) affected

0.000 sec

Query 12: -SELECT

```
SELECT *
FROM customers_transactions;
```

Output: -

60 15:33:32 SELECT \* FROM customers\_transactions LIMIT 0, 1000 10 row(s) returned 0.016 sec / 0.000 sec

Result Grid

Filter Rows:

Edit:

Export/Import:

|  | txn_id | customer_name        | branch_name | transaction_type | amount   | transaction_date |
|--|--------|----------------------|-------------|------------------|----------|------------------|
|  | 101    | G. Hemanth Venkatesh | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
|  | 102    | G. Hemanth Venkatesh | Hyderabad   | Withdrawal       | 2000.00  | 2024-01-06       |
|  | 103    | Akshara Reddy        | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|  | 104    | Anirudh Reddy        | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
|  | 105    | Yashwanth            | Hyderabad   | Withdrawal       | 5000.00  | 2024-01-11       |
|  | 106    | Adithya Vardhan      | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|  | 107    | Ananaya Varshitha    | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       |
|  | 108    | Varnika Guptha       | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
|  | 109    | Prerka               | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       |
|  | 110    | Nihita               | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|  | NULL   | NULL                 | NULL        | NULL             | NULL     | NULL             |

Query 13: -SELECT WITH WHERE

```
SELECT *
FROM customers_transactions
WHERE transaction_type='Deposit';
```

Output: -

62 15:35:53 SELECT \* FROM customers\_transactions WHERE transaction\_type = 'Deposit' 6 row(s) returned 0.000 sec / 0.000 sec

Result Grid

Filter Rows:

Edit:

Export/Import:

|   | txn_id | customer_name        | branch_name | transaction_type | amount   | transaction_date |
|---|--------|----------------------|-------------|------------------|----------|------------------|
|   | 101    | G. Hemanth Venkatesh | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
|   | 103    | Akshara Reddy        | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|   | 104    | Anirudh Reddy        | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
|   | 106    | Adithya Vardhan      | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|   | 108    | Varnika Guptha       | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
|   | 110    | Nihita               | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
| * | NULL   | NULL                 | NULL        | NULL             | NULL     | NULL             |

|   | txn_id | customer_name        | branch_name | transaction_type | amount   | transaction_date |
|---|--------|----------------------|-------------|------------------|----------|------------------|
| ▶ | 101    | G. Hemanth Venkatesh | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
|   | 103    | Akshara Reddy        | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|   | 104    | Anirudh Reddy        | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
|   | 106    | Adithya Vardhan      | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|   | 108    | Varnika Guptha       | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
|   | 110    | Nihita               | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|   | NULL   | NULL                 | NULL        | NULL             | NULL     | NULL             |

Query 14: -SELECT WITH ORDER BY

```
SELECT *
FROM customers_transactions
ORDER BY amount DESC;
```

Output: -

64

15:37:59

SELECT \* FROM customers\_transactions ORDER BY amo...

10 row(s) returned

0.016 sec / 0.000 sec

Result Grid

Filter Rows:

Edit:

Export/Import:

|   | txn_id | customer_name        | branch_name | transaction_type | amount   | transaction_date |
|---|--------|----------------------|-------------|------------------|----------|------------------|
| ▶ | 106    | Adithya Vardhan      | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|   | 103    | Akshara Reddy        | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|   | 110    | Nihita               | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|   | 108    | Varnika Guptha       | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       |
|   | 104    | Anirudh Reddy        | Vizag       | Deposit          | 8000.00  | 2024-01-10       |
|   | 101    | G. Hemanth Venkatesh | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       |
|   | 105    | Yashwanth            | Hyderabad   | Withdrawal       | 5000.00  | 2024-01-11       |
|   | 109    | Prerka               | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       |
|   | 102    | G. Hemanth Venkatesh | Hyderabad   | Withdrawal       | 2000.00  | 2024-01-06       |
|   | 107    | Ananaya Varshitha    | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       |
| ● | NULL   | NULL                 | NULL        | NULL             | NULL     | NULL             |

Query 15: - DCL (DATA CONTROL LANGUAGE)

GRANT ALL

```
GRANT ALL PRIVILEGES
ON ddl_db.customers_transactions
TO 'BranchManager'@'localhost';
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Output: -

|   |    |          |                                                                                      |                   |           |
|---|----|----------|--------------------------------------------------------------------------------------|-------------------|-----------|
| ✓ | 72 | 15:48:57 | GRANT ALL PRIVILEGES ON ddl_db.customers_transactions TO 'BranchManager'@'localhost' | 0 row(s) affected | 0.000 sec |
|---|----|----------|--------------------------------------------------------------------------------------|-------------------|-----------|

Query 16: -REVOKE ALL

```
REVOKE ALL PRIVILEGES
ON ddl_db.customers_transactions
FROM 'BranchManager'@'localhost';
```

Output: -

|   |    |          |                                                                                         |                   |           |
|---|----|----------|-----------------------------------------------------------------------------------------|-------------------|-----------|
| ✓ | 74 | 15:49:33 | REVOKE ALL PRIVILEGES ON ddl_db.customers_transactions FROM 'BranchManager'@'localhost' | 0 row(s) affected | 0.000 sec |
|---|----|----------|-----------------------------------------------------------------------------------------|-------------------|-----------|

Query 17: -TCL (TRANSACTION CONTROL LANGUAGE)

COMMIT

```
UPDATE customer_transactions
SET amount=6000
WHERE txn_id=101;

COMMIT;
```

Output: -

|   |    |          |                                                               |                   |                 |            |             |           |
|---|----|----------|---------------------------------------------------------------|-------------------|-----------------|------------|-------------|-----------|
| ✓ | 76 | 15:52:18 | UPDATE customer_transactions SET amount=6000 WHERE txn_id=101 | 1 row(s) affected | Rows matched: 1 | Changed: 1 | Warnings: 0 | 0.000 sec |
| ✓ | 77 | 15:52:25 | COMMIT                                                        | 0 row(s) affected | 0.000 sec       |            |             |           |

Query 18: - SAVEPOINT

```
-- 2520030071-G. Hemanth Venkatesh
SAVEPOINT Before_Update;
```

Output: -

|   |    |          |                         |                   |           |
|---|----|----------|-------------------------|-------------------|-----------|
| ✓ | 78 | 15:53:32 | SAVEPOINT Before_Update | 0 row(s) affected | 0.000 sec |
|---|----|----------|-------------------------|-------------------|-----------|

Query 19: -ROLLBACK TO SAVEPOINT

```
-- 2520030071-G. Hemanth Venkatesh
UPDATE customer_transactions
SET amount=99999
WHERE txn_id=102;
ROLLBACK TO Before_Update;
```

Output: -

|   |    |          |                                                                |                   |                 |            |             |           |
|---|----|----------|----------------------------------------------------------------|-------------------|-----------------|------------|-------------|-----------|
| ✓ | 84 | 15:55:57 | UPDATE customer_transactions SET amount=99999 WHERE txn_id=102 | 0 row(s) affected | Rows matched: 0 | Changed: 0 | Warnings: 0 | 0.000 sec |
|---|----|----------|----------------------------------------------------------------|-------------------|-----------------|------------|-------------|-----------|